

Ritali Ghosh

Curriculum Vitae

Ph.D., Astronomy and Astrophysics

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 [RitaliG](#)

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I am a computational astrophysicist studying the dynamics of multiphase gas in galaxy evolution. My work uses high-resolution numerical simulations to investigate turbulence, mixing, and radiative cooling in galactic outflows and ram-pressure stripped galaxies. During my PhD at the Indian Institute of Science, I developed simulation models of cold-gas survival and growth in hot galactic winds, combining hydrodynamic simulations with theoretical modelling.

Experience

2025 Oct – present	Research Fellow (Bridge Position) working with Dr. Max Gronke (ASGARD Group at ARI/ZAH)	Astronomisches Rechen-Institut, University of Heidelberg Baden-Wuerttemberg, Germany
2025 Mar – mid Oct	Long-term Visiting Fellow	Max Planck Institute for Astrophysics Garching, Germany
2019 Aug – 2025 Jul	Graduate Student Supervisor: Prof. Prateek Sharma Thesis: Multiphase gas in the baryon cycle - from jellyfish galaxies to turbulent galactic outflows CGPA: 8.9 on a 10-point scale in the Joint Astronomy & Astro-physics Program (JAP) coursework	Indian Institute of Science Bengaluru, India

Education

2015 Jun – 2017 May	Master of Science, Physics CGPA: 4.69 on a scale of 6.0 (Grade: A+) Specialization: Particle Physics & Quantum Field Theory DST-INSPIRE Scholar; ranked 5th in the batch	University of Calcutta Kolkata, India
2012 Apr – 2015 May	Bachelor of Science, Physics CGPA: 7.1 on a 10-point scale (Grade: A) DST-INSPIRE Scholar; awarded the “Chameli Bose Memorial Prize” for highest proficiency in B.Sc. Part II from Bethune College	Bethune College Kolkata, India

Publications

4.

Woven by the Whirls: The growth and entrainment of cold clouds in turbulent hot winds
under review in the *Monthly Notices of the Royal Astronomical Society* ([arxiv:2510.03552](#))
[Ritali Ghosh](#), Max Gronke, Prateek Sharma and Alankar Dutta

3.

Jellyfish galaxies in clusters: the circumgalactic contribution to the cold gas in the stripped tails
(in prep; draft available upon request)
[Ritali Ghosh](#), Alankar Dutta and Prateek Sharma

2.

Beyond profiles: Using log-normal distributions to model the multiphase circumgalactic medium
Monthly Notices of the Royal Astronomical Society, Volume 531, Issue 4, July 2024, Pages 5117–5139
(doi:[10.1093/mnras/stae977](#))
Alankar Dutta, Mukesh Singh Bisht, Prateek Sharma, [Ritali Ghosh](#), Manami Roy and Biman B. Nath

1.

Ram pressure stripping in clusters: Gravity can bind the ISM but not the CGM
Monthly Notices of the Royal Astronomical Society, Volume 531, Issue 3, July 2024, Pages 3445–3467
(doi:[10.1093/mnras/stae1345](#))
[Ritali Ghosh](#), Alankar Dutta, Prateek Sharma

Technical & Language Skills

- Hydrodynamical simulations using **PLUTO**, **Athena++**, **Arepo**
- Proficient in **High Performance Computing (HPC)** and Parallel Programming with **C/C++**
- Extensively used **ParaView** for large-scale data visualization and analysis; contributed to **Catalyst** integration for *in-situ* visualization with PLUTO
- Data analysis using **Python** (with yt, numpy, matplotlib, scipy, h5py, mpi4py, scikit-image, cupy)
- Executed large-scale HPC simulations, utilizing approximately 3 million CPU hours during the PhD
- Languages: English (proficient), Hindi (native speaker), Bengali (conversational), German (A1 level)

Talks and Research Visits

Conference Talks

Modelling of Multiphase Astrophysical Media, Ringberg, Germany	Nov 2025
7th ICM Theory & Computation Workshop, 2024 (<i>online</i>)	Jun 2024
Observe Local Think Global: What Solar Observations can teach us about Multiphase Plasmas across Astrophysical Scales, 2024, Bern, Switzerland	May 2024
Astronomical Society of India Meeting, ASI 2024, Bangalore, India	Feb 2024
Modelling of Multiphase Astrophysical Media, Aspenstein Castle at Lake Kochel, Germany	Jun 2023
GALFLOWS 2023: A National Workshop on Galactic inflows and outflows on all Scales, Pune, India	Feb 2023
What matter(s) around galaxies 2022, Champoluc, Italy	Sep 2022

Seminar Talks

ARI Institute Colloquium, University of Heidelberg	8 Jan 2026
ORIGINS Turbulence Day, Max Planck institute for Astrophysics, Garching	22 Jul 2025
Leibniz-Institut für Astrophysik, Potsdam	6 Nov 2024
Institute for Theoretical Astrophysics (ITA) Heidelberg	4 Nov 2024
CGM Meeting, European Southern Observatory, Garching	25 Oct 2024
MPIA Galaxy Coffee, Heidelberg	17 Oct 2024
Weekly CAST Meeting, Universitäts-Sternwarte München, Munich	17 Nov 2023
CGM Meeting, European Southern Observatory, Garching	15 Nov 2023
In-house Symposium, Department of Physics, IISc, Bengaluru (<i>online</i>)	10 Nov 2023
MPA Galaxy Group Meeting, Max Planck Institute for Astrophysics, Garching	7 Jun 2023
MPA Cosmology Seminar, Max Planck Institute for Astrophysics, Garching,	20 Sep 2022
Dipartimento di Fisica e Astronomia, INAF, Italy	19 Sep 2022

Poster & Flash Talks

Poster Presentation, 18th Potsdam Thinkshop 2025, Potsdam	Jul 2025
Flash Talk + Poster presentation, Baryons Beyond Galactic Boundaries 2024, Pune, India	Dec 2024
Poster Presentation, Astronomical Society of India Meeting, ASI 2022, IIT Roorkee	Mar 2022
KITP Program: Fundamentals of Gaseous Halos (<i>online</i> attendee)	Jan 2022

Academic Awards

National Level Qualified Examinations

- Qualified GATE Fellowship 2019
- Qualified UGC-JRF NET December 2018
- Qualified JEST 2019

Academic Achievements and Scholarships

- Ranked 5th in University of Calcutta, M.Sc. Batch 2015-17
- DST-INSPIRE Scholarship 2012-2017
- Mamraj Agarwal Rashtriya Puraskar, 2012 (*felicitated for scoring 96.6% in CBSE Class XII Exams, 2012 by the then Governor of West Bengal, India*)
- Certificate of Merit in Mathematics 2012 (*for being in top 0.1% of All India Senior School Certificate Exam, 2012 in Mathematics; Score: 100%*)

Teaching & Mentoring Experience

Co-supervision of Master student - Jiazhou Xu, visiting student at Astronomisches Rechen-Institut, Heidelberg
October — December 2025

Teaching Assistantship — Galaxies and the Interstellar Medium

January — May 2023

assisted Prof. Prateek Sharma and Prof. Biman B. Nath in the PH 365 course for Joint Astrophysics Program, IISc

Teaching Assistantship — Introduction to Fluid Mechanics and Plasma Physics

August — December 2021

assisted Prof. Arnab Rai Choudhuri in the PH 363 course for Joint Astrophysics Program, IISc

Professional activities and outreach

Reviewer, New Astronomy (ongoing)

Hands-on Sessions, Computational MHD with PLUTO (May 2023)

Organized hands-on sessions on basics of MPI parallel computation for about 60 beginning Ph.D. and advanced master's students in the Workshop on Computational MHD with the PLUTO code at IISc, Bangalore, India

Coordinator, Weekly arXiv Meetings (2021-22; 2022-23)

Coordinated the weekly arXiv meetings for the Astrophysics Group at the Department of Physics, IISc from 2021 to 2023

Volunteer, IISc Open Day (Feb 2020, Mar 2023, Feb 2024)

Actively participated in popular science lectures and hands-on demonstrations to around 50,000 people at the annual Open Day event at the IISc, Bangalore, India

Astronomy Outreach Volunteer, International Astronomical Union (2019)

Engaged with approximately 600 primary and high school students at Kendriya Vidyalaya IISc, Bangalore, India as a part of the International Astronomical Union (IAU) global project 'Astronomy Day in school'

Professional References

Prof. Prateek Sharma, Indian Institute of Science, Bengaluru, India

✉ prateek@iisc.ac.in

Dr. Max Gronke, ARI/ZAH, University of Heidelberg, Germany

✉ max.gronke@uni-heidelberg.de

Prof. Biman B. Nath, Raman Research Institute, Bengaluru, India

✉ nath.biman@gmail.com